

■ PDF Hash Extractor

A Python GUI tool to compute cryptographic file hashes of PDF documents

Python 3.8+

Tkinter GUI

No Dependencies

Cross-Platform

Overview

PDF Hash Extractor is a lightweight desktop application built with Python's built-in **tkinter** library. It allows users to select any PDF file and instantly compute five industry-standard cryptographic hash values. No third-party packages are required — everything runs on the Python standard library.

Use cases include:

- **File Integrity Verification:** Confirm a PDF has not been tampered with after download or transfer.
- **Forensic Analysis:** Record cryptographic fingerprints of PDF evidence files.
- **Duplicate Detection:** Identify identical files regardless of filename.
- **Security Auditing:** Log hash values for document provenance tracking.

Features

Feature	Details
Hash Algorithms	MD5, SHA-1, SHA-256, SHA-512, SHA3-256
GUI Framework	tkinter (built into Python — zero installs)
File Info	Displays filename and file size in bytes
Clipboard Export	Copy all hashes to clipboard with one click
Scrollable Output	Handles long hash strings cleanly
Cross-Platform	Windows, macOS, Linux

Installation & Requirements

Requirements: Python 3.8 or higher. No pip installs needed.

1. Clone the repository:

```
git clone https://github.com/yourusername/pdf-hash-extractor.git
cd pdf-hash-extractor
```

2. Run the application:

```
python pdf_hash_extractor.py
```

3. (Optional) Build a standalone EXE with PyInstaller:

```
pip install pyinstaller
pyinstaller --onefile --windowed --name PDFHashExtractor pdf_hash_extractor.py
```

The compiled .exe will appear in the **dist/** folder.

How to Use

1	Launch Run pdf_hash_extractor.py or the compiled .exe.
2	Select PDF Click the '■ Select PDF' button and browse to your PDF file.
3	View Hashes All five hash values are computed instantly and displayed in the output panel.
4	Copy Click '■ Copy to Clipboard' to copy all results for use elsewhere.

Hash Algorithms Reference

Algorithm	Output Length	Use Case	Speed
MD5	128-bit / 32 hex	Quick integrity check (not secure)	Fastest
SHA-1	160-bit / 40 hex	Legacy systems (deprecated for security)	Fast
SHA-256	256-bit / 64 hex	General purpose — recommended standard	Medium
SHA-512	512-bit / 128 hex	High-security applications	Medium

SHA3-256	256-bit / 64 hex	Modern standard (Keccak-based)	Medium
----------	------------------	--------------------------------	--------

Core Hash Extraction Logic

The hash computation at the heart of the tool:

```
import hashlib

def extract_hashes(path):
    with open(path, "rb") as f:
        data = f.read()

    return {
        "MD5":      hashlib.md5(data).hexdigest(),
        "SHA-1":    hashlib.sha1(data).hexdigest(),
        "SHA-256":  hashlib.sha256(data).hexdigest(),
        "SHA-512":  hashlib.sha512(data).hexdigest(),
        "SHA3-256": hashlib.sha3_256(data).hexdigest(),
    }
```

Repository Structure

```
pdf-hash-extractor/
■■■ pdf_hash_extractor.py    # Main application
■■■ README.md               # Project readme
■■■ LICENSE                 # MIT License
■■■ requirements.txt        # Empty – stdlib only
■■■ docs/
■   ■■■ PDF_Hash_Extractor_Docs.pdf
■■■ dist/
    ■■■ PDFHashExtractor.exe # (after PyInstaller build)
```

License

This project is released under the **MIT License**. You are free to use, modify, and distribute this software for personal or commercial purposes with attribution.